

Business Insights

Supply Chain Management & Logistics Analysis

The SQL analysis and Power BI dashboard convert the supply chain dataset into measurable procurement, supplier, quality, freight, and delivery findings.

1. Procurement Overview

The dashboard shows approximately **₹991.94M in actual procurement spend across 480 purchase orders**.

Monthly and category-level actual-vs-target analysis provides visibility into where procurement spending differs from plan.

2. Purchase-Price Variance

Average purchase-price variance is **1.12%**.

Supplier-level analysis shows clear differences in PPV, with the highest displayed supplier at **2.68%**. This identifies supplier-level price variance that can be reviewed against contracted pricing.

3. Supplier Performance

The dashboard contains **48 suppliers**.

Key supplier KPIs shown are:

- Average supplier rating: **4.09**
- Average quality score: **87.10**
- Average delivery score: **83.31**

The supplier performance matrix helps identify the relationship between supplier lead time and delivery performance.

4. Supplier Risk

The dashboard shows:

- **35 Medium-risk suppliers (72.92%)**
- **13 Low-risk suppliers (27.08%)**

No High-risk supplier segment is displayed in the current dashboard view.

This gives procurement teams a supplier-risk monitoring baseline while the supplier performance matrix identifies operational concerns that may require review.

5. Delivery Reliability

Overall on-time delivery is **63.71%**.

The monthly OTD trend shows:

- **69.0% in January**
- **68.9% in February**
- **65.8% in March**
- **62.6% in April**
- **67.9% in May**
- **66.7% in June**

- **63.4% in July**
- **62.1% in August**
- **60.2% in September**
- **58.5% in October**
- **58.9% in November**
- **59.3% in December**

The trend shows a clear deterioration during most of the year, followed by a small recovery in the final two months.

6. Delivery Delay Analysis

The dashboard records **487 delayed shipments**.

The visible delay breakdown is:

- Road: **302**
- Rail: **113**
- Air: **72**

Road therefore represents the largest visible source of delayed shipments.

The dashboard also shows an overall **delivery SLA gap of -0.27**.

7. Carrier Performance

Displayed carrier OTD percentages range from:

- GATI Industrial: **57.89%**
- RailCargo Connect: **59.17%**
- TCI Express: **60.09%**
- Safexpress: **62.41%**
- VRL Logistics: **63.33%**
- SwiftHaul Logistics: **67.59%**
- BlueDart Freight: **71.43%**
- Delhivery B2B: **72.34%**

This shows significant variation in carrier delivery reliability and supports carrier-level SLA monitoring.

8. Freight Cost

Total freight cost is approximately **₹24.13M**.

By transport mode:

- Air: **₹12.44M (51.54%)**
- Road: **₹9.02M (37.39%)**
- Rail: **₹2.67M (11.08%)**

Air has the largest share of freight cost in the dashboard.

9. Transit Time

Overall average transit time is **7.96 days**.

Displayed carrier averages range from approximately **7.4 days to 8.3 days**.

This demonstrates that transit performance differs across carriers and can be analyzed together with freight cost and OTD.

10. Quality and Rejection

The overall rejection rate displayed on the dashboard is **0.82%**.

The category analysis highlights individual product categories with higher operational rejection or delay counts, creating areas for further supplier-quality investigation.

11. Purchase Order Status

The dashboard shows:

- **93.57% Closed**
- **5.26% Cancelled**

The remaining purchase orders are in other/open statuses represented in the dashboard.

This provides visibility into the current PO pipeline and status mix.

12. Regional Delivery

The regional OTD chart shows the largest visible volumes of on-time deliveries from:

- West: **295**
- South: **253**
- North: **162**
- Central: **83**
- East: **62**

The regional view helps identify where delivery activity and on-time performance are concentrated.

Final Business Outcome

The project addresses the main supply chain problems by identifying:

1. Procurement spend variance
2. Purchase-price variance and potential price leakage
3. Supplier performance differences
4. Supplier risk exposure
5. Delivery reliability deterioration
6. Carrier SLA gaps
7. Freight-cost concentration by transport mode
8. Transit-time variation
9. Quality and rejection issues
10. Purchase-order status patterns
11. Regional and transport-mode delivery patterns

These findings support practical business actions such as supplier price reviews, supplier performance monitoring, carrier SLA review, road-shipment delay investigation, freight-mode cost analysis, quality monitoring, and procurement target tracking.

Overall Outcome

The project demonstrates a complete analytics workflow:

Excel Dataset



Data Preparation



MySQL Database



SQL Business Analysis



Power BI Dashboard



Business Findings